Team Members:

Roll Numbers	Names	Roles
22i-2327	Muhammad Rohaan	Developer
22i-2366	Ammaar Ahmed	Analyst + Designer
23i-0721	Muhammad Farzeen Khan Tareen	Team Manager + QA Manager

Company Name:

TerraNex Technologies

Company Logo:



National University of Computer and Emerging Sciences, Islamabad
Department of Computer Science



Inventory & Stock Management System

Project Team

Muhammad Rohaan (22i-2327)

Ammar Ahmed (22i-2366)

Muhammad Farzeen Khan Tareen (23i-0721)

Inventory & Stock Management System

1. Project Overview

This project is about creating a computer-based system to manage and track stock in an organization. The system will let users add new products, update details, and remove items from stock. It will also keep a record of how much stock is available in the organization, give alerts when stock is low in the organization, and store supplier information. Because while doing manual work people do a lot of mistakes. This system will help to reduce mistake and make inventory handling more faster and more accurate.

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2. Motivation

Many local and small businesses still use manual methods like notebooks or spreadsheets to manage inventory. This often causes problems for them like having too much stock, running out of items, or losing track of products. A proper system like this will help keep records safe, updated, and easy to access for these businesses. The main reason for making this project is to help businesses to save time, avoid mistakes, and make stock management simpler and more reliable.

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3. Objectives

The goals of this project are:

- To design a desktop application using JavaFX.
- To store and manage inventory data in a database using SQL.
- To add, update, or delete product details easily.
- To show alerts when stock is low.
- To keep supplier details linked with products.
- To generate reports like monthly stock use and purchase history.

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4. Expected Outcomes

After finishing this project, we expect to have:

- A simple and easy-to-use desktop system for managing inventory.

- Secure login for admin and staff.
- Real-time stock updates with alerts for low stock.
- Automatic reports for stock, suppliers, and product movement.
- A system that reduces manual work and improves accuracy.

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5. Project Scope

The system will include these features:

- **Product Management**: Add, update, delete, and search products.
- **Stock Management**: Record stock coming in and going out.
- **Supplier Management**: Store supplier details.
- **Alerts**: Show messages when stock is low.
- **Reports**: Create stock summary and supplier reports.

The system will not include online access, barcode scanning, or multiple branches. It will focus only on a single organization's stock management.

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6. Tools and Technologies

- **Frontend**: JavaFX for the application interface.
- **Backend Database**: SQL.
- **Development Environment**: IntelliJ or Eclipse.

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Fully Dressed Use cases

Use Case 1: User Login

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>User Login</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin/Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin/Staff: wants secure system access.
Admin: ensures authorized users only.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>User launches application.</td><td>System displays login screen.</td></tr>
<tr><td>User enters username/password and clicks Login.</td><td>System validates credentials.</td></tr>
<tr><td>Credentials valid.</td><td>System grants access to appropriate dashboard.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Invalid credentials → system shows 'Invalid username or password'.</td></tr>
<tr><td>Preconditions</td><td>User must be registered with valid credentials.</td></tr>
<tr><td>Postconditions</td><td>User successfully logged in.</td></tr>
</table>

Use Case 2: Add New Product

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Add New Product</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: maintains product catalog.
Staff: uses products for stock tracking.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin opens Product Management > Add Product.</td><td>System shows product entry form.</td></tr>
<tr><td>Admin enters product details and submits.</td><td>System validates and saves product.</td></tr>
<tr><td>Valid details provided.</td><td>System confirms successful addition.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Missing fields → highlight and prompt correction.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in; product not already exists.</td></tr>
<tr><td>Postconditions</td><td>New product added to database.</td></tr>
</table>

Use Case 3: Update Product Details

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Update Product Details</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: maintains accurate and updated product info.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin searches for product.</td><td>System shows product details.</td></tr>
<tr><td>Admin edits and saves changes.</td><td>System validates and updates database.</td></tr>
<tr><td>Changes valid.</td><td>System confirms update.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Invalid data → system rejects changes.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in and product exists.</td></tr>
<tr><td>Postconditions</td><td>Product details updated.</td></tr>
</table>

Use Case 4: Record Stock Inflow

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Record Stock Inflow</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Staff: records received goods.
Admin: keeps accurate total stock.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Staff selects 'Stock Inflow' option.</td><td>System shows inflow form.</td></tr>
<tr><td>Staff enters product and quantity.</td><td>System validates entry and updates stock.</td></tr>
<tr><td>Entry valid.</td><td>System confirms addition and logs transaction.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Quantity invalid → show error.</td></tr>
<tr><td>Preconditions</td><td>Staff logged in, product exists.</td></tr>
<tr><td>Postconditions</td><td>Stock quantity increased, transaction logged.</td></tr>
</table>

Use Case 5: Process Stock Outflow

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Process Stock Outflow</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Staff: records sales or damaged goods.
Admin: monitors stock deductions.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Staff selects 'Stock Outflow'.</td><td>System shows outflow form.</td></tr>
<tr><td>Staff enters product, quantity, and reason.</td><td>System validates and deducts stock.</td></tr>
<tr><td>Sufficient stock available.</td><td>System confirms deduction and logs event.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Insufficient stock → system shows 'Not enough quantity'.</td></tr>
<tr><td>Preconditions</td><td>Staff logged in, sufficient stock available.</td></tr>
<tr><td>Postconditions</td><td>Stock decreased, transaction recorded.</td></tr>
</table>

Use Case 6: Check Low Stock Alerts

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Check Low Stock Alerts</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin/Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Staff/Admin: monitors low-stock items to reorder timely.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>User opens dashboard.</td><td>System checks all stock levels.</td></tr>
<tr><td>Threshold crossed for any item.</td><td>System lists items in 'Low Stock Alerts'.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>No low-stock items → show 'All stock levels sufficient'.</td></tr>
<tr><td>Preconditions</td><td>User logged in, products have thresholds.</td></tr>
<tr><td>Postconditions</td><td>Low-stock items displayed on dashboard.</td></tr>
</table>

Use Case 7: Add New Supplier

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Add New Supplier</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: maintains supplier database for product linking.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin navigates to Supplier Management.</td><td>System shows add supplier form.</td></tr>
<tr><td>Admin fills supplier details and submits.</td><td>System validates and saves record.</td></tr>
<tr><td>All valid.</td><td>System confirms supplier added.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Missing data → prompt correction.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in; supplier not duplicate.</td></tr>
<tr><td>Postconditions</td><td>New supplier added to system.</td></tr>
</table>

Use Case 8: Generate Stock Summary Report

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Generate Stock Summary Report</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: wants overview of stock for decisions.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin selects 'Stock Summary Report'.</td><td>System retrieves stock data.</td></tr>
<tr><td>Admin requests display.</td><td>System generates and displays summary.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>No data → show 'No records available'.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in.</td></tr>
<tr><td>Postconditions</td><td>Report displayed on screen.</td></tr>
</table>

Use Case 9: Generate Purchase History Report

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Generate Purchase History Report</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: analyzes inflow over time for trends.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin selects 'Purchase History Report'.</td><td>System prompts date range.</td></tr>
<tr><td>Admin enters range.</td><td>System compiles inflow transactions and displays report.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Invalid date range → system prompts correction.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in.</td></tr>
<tr><td>Postconditions</td><td>Purchase report generated and displayed.</td></tr>
</table>

Use Case 10: Search for a Product

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Search for a Product</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin/Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin/Staff: quickly find product details.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>User enters search keyword.</td><td>System queries database.</td></tr>
<tr><td>Matches found.</td><td>System displays results list.</td></tr>
</table></td></tr>
<tr><td>Extensions</td><td>No matches → system shows 'No products found'.</td></tr>
<tr><td>Preconditions</td><td>User logged in.</td></tr>
<tr><td>Postconditions</td><td>Matching products displayed.</td></tr>
</table>

Use Case 11: Manage User Accounts

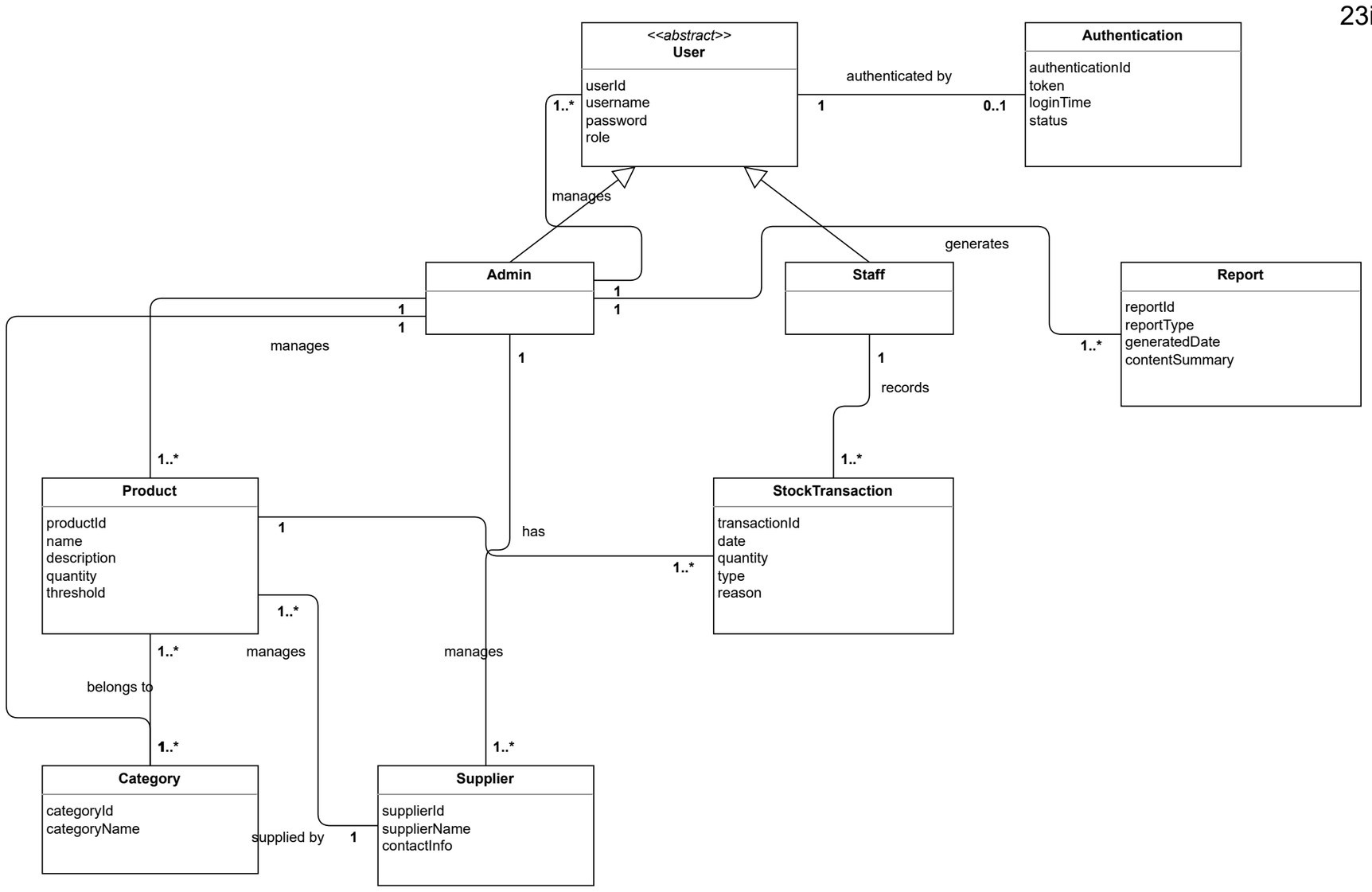
<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Manage User Accounts</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>Summary</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: manages system access and permissions.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin opens User Management.</td><td>System lists all users.</td></tr>
<tr><td>Admin creates/updates/deactivates user.</td><td>System validates and saves changes.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Admin attempts to deactivate self → show 'Action not allowed'.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in.</td></tr>
<tr><td>Postconditions</td><td>User accounts updated.</td></tr>
</table>

Use Case 12: Manage Product Categories

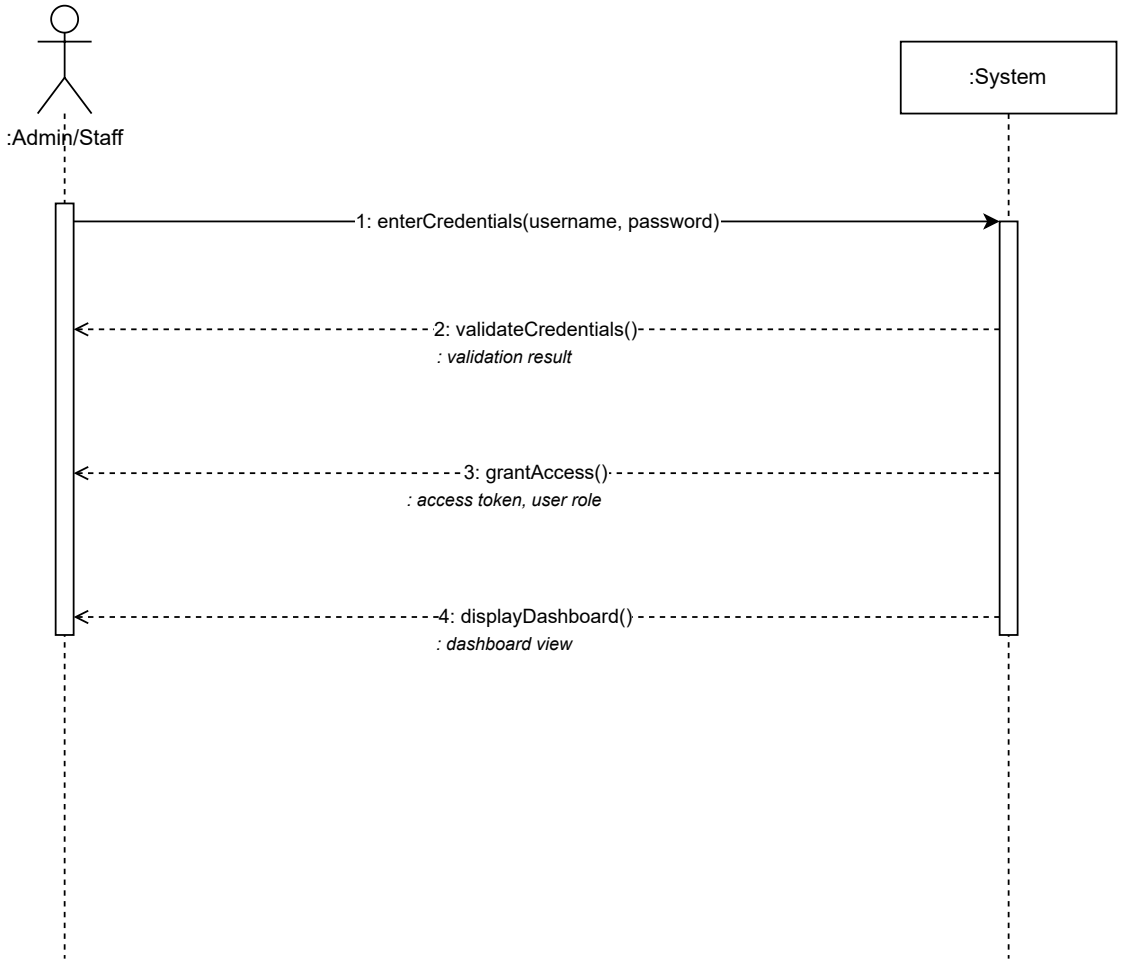
<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Manage Product Categories</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>Summary</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: organizes and maintains product categories.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin opens Category Management.</td><td>System lists categories.</td></tr>
<tr><td>Admin adds, edits, or deletes category.</td><td>System validates and updates database.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Category in use → prompt reassignment before deletion.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in.</td></tr>
<tr><td>Postconditions</td><td>Categories added/edited/deleted.</td></tr>
</table>

Inventory & Stock Management System – Domain Model Diagram

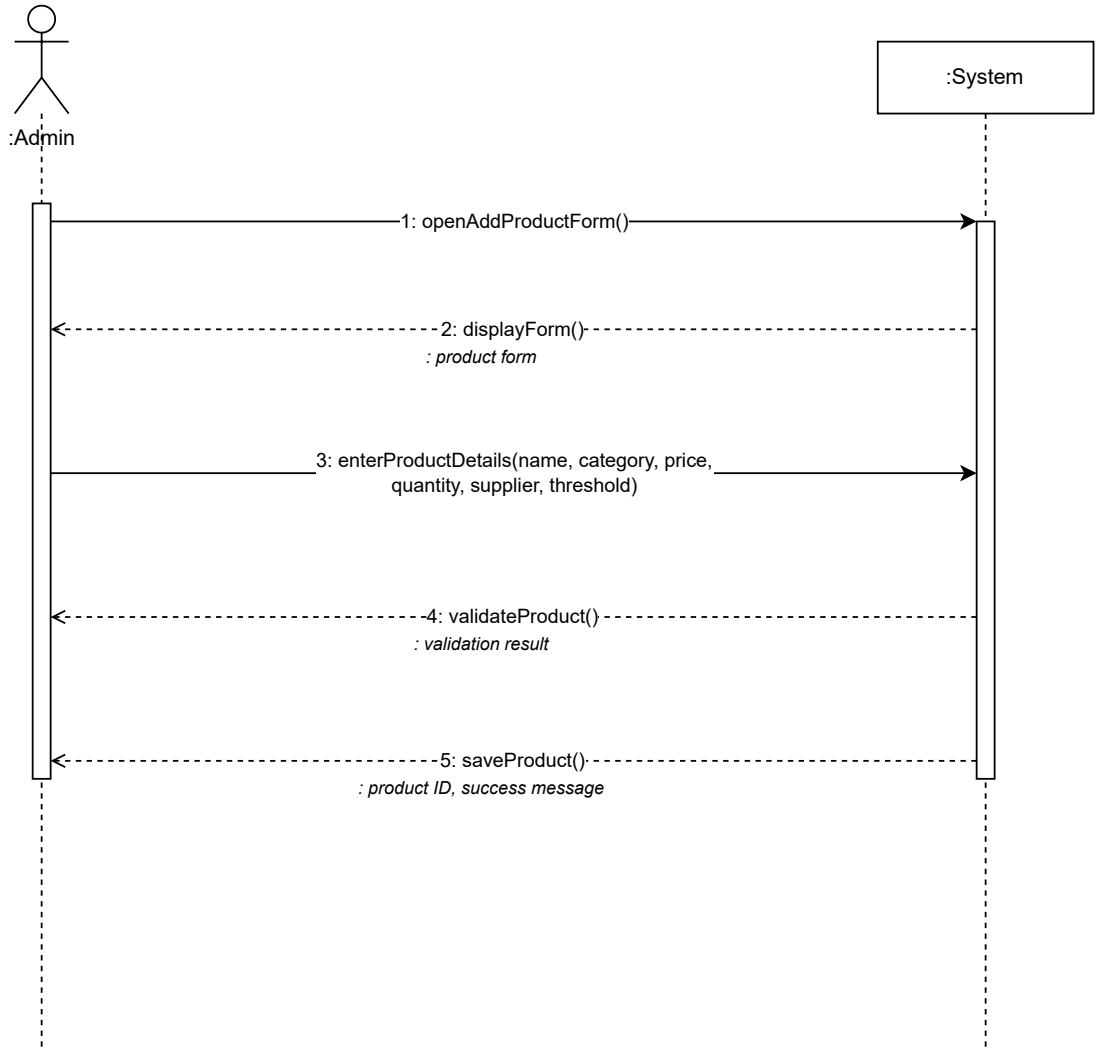
22i-2327 M.Rohaan
22i-2366 M.Ammar
23i-0721 Farzeen Tareen



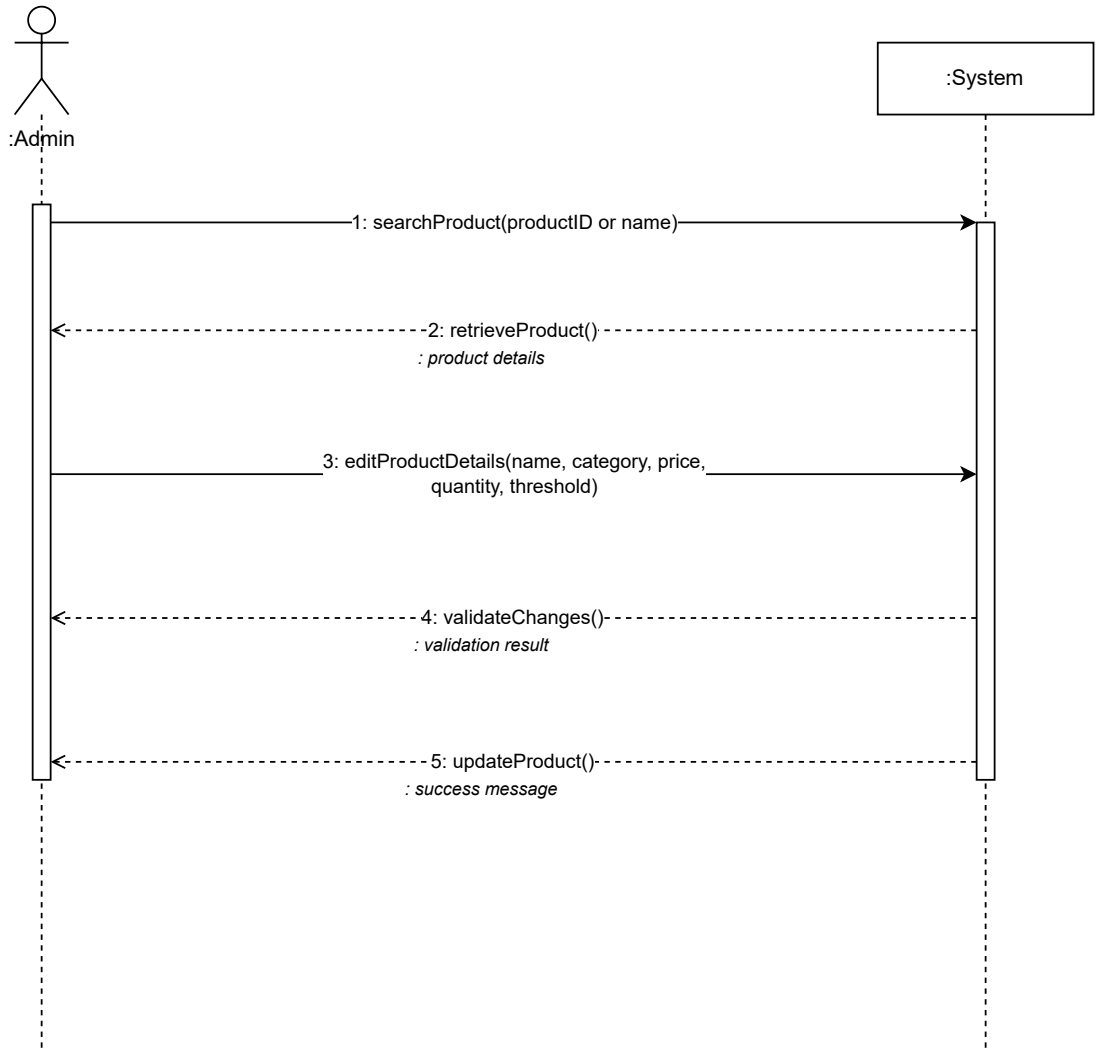
SSD - User Login



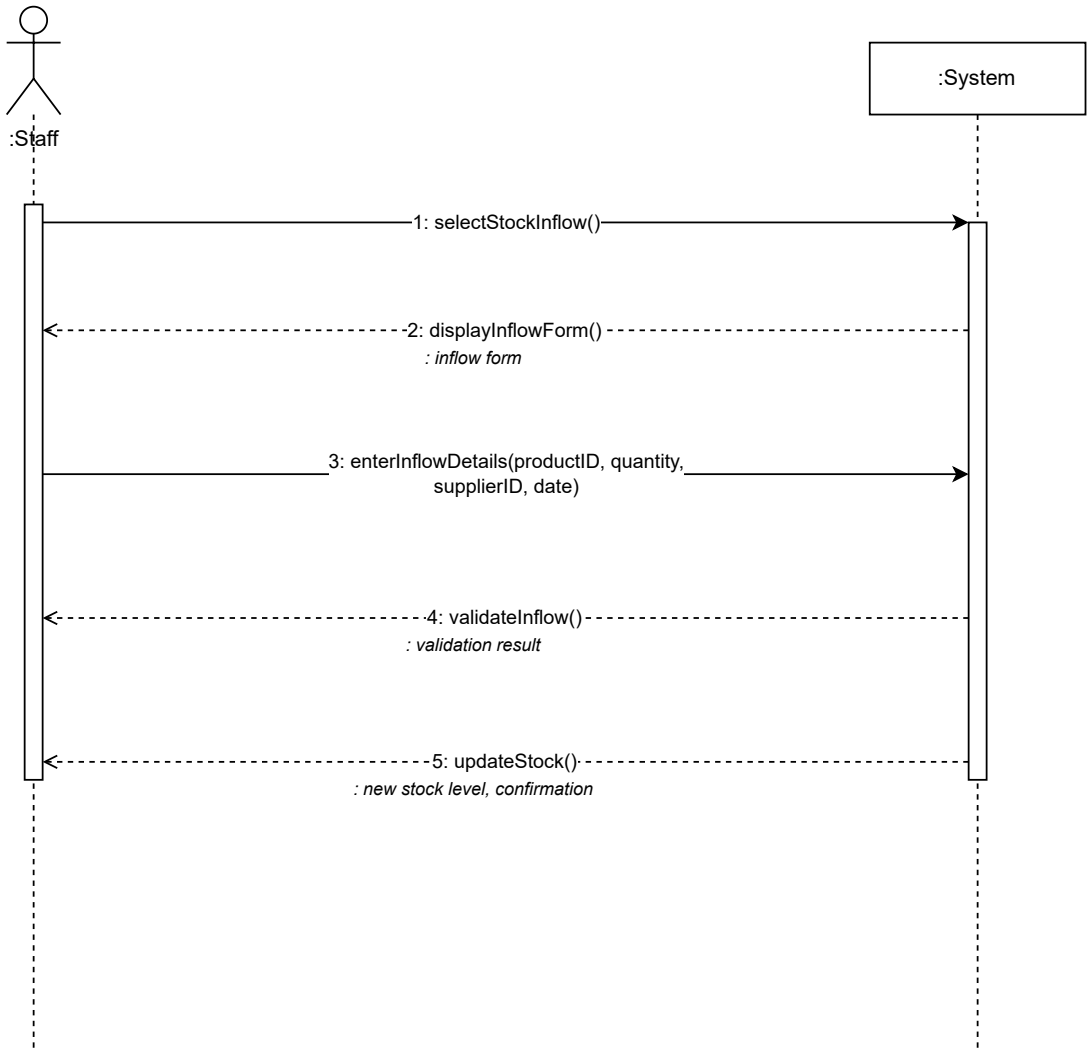
SSD - Add New Product



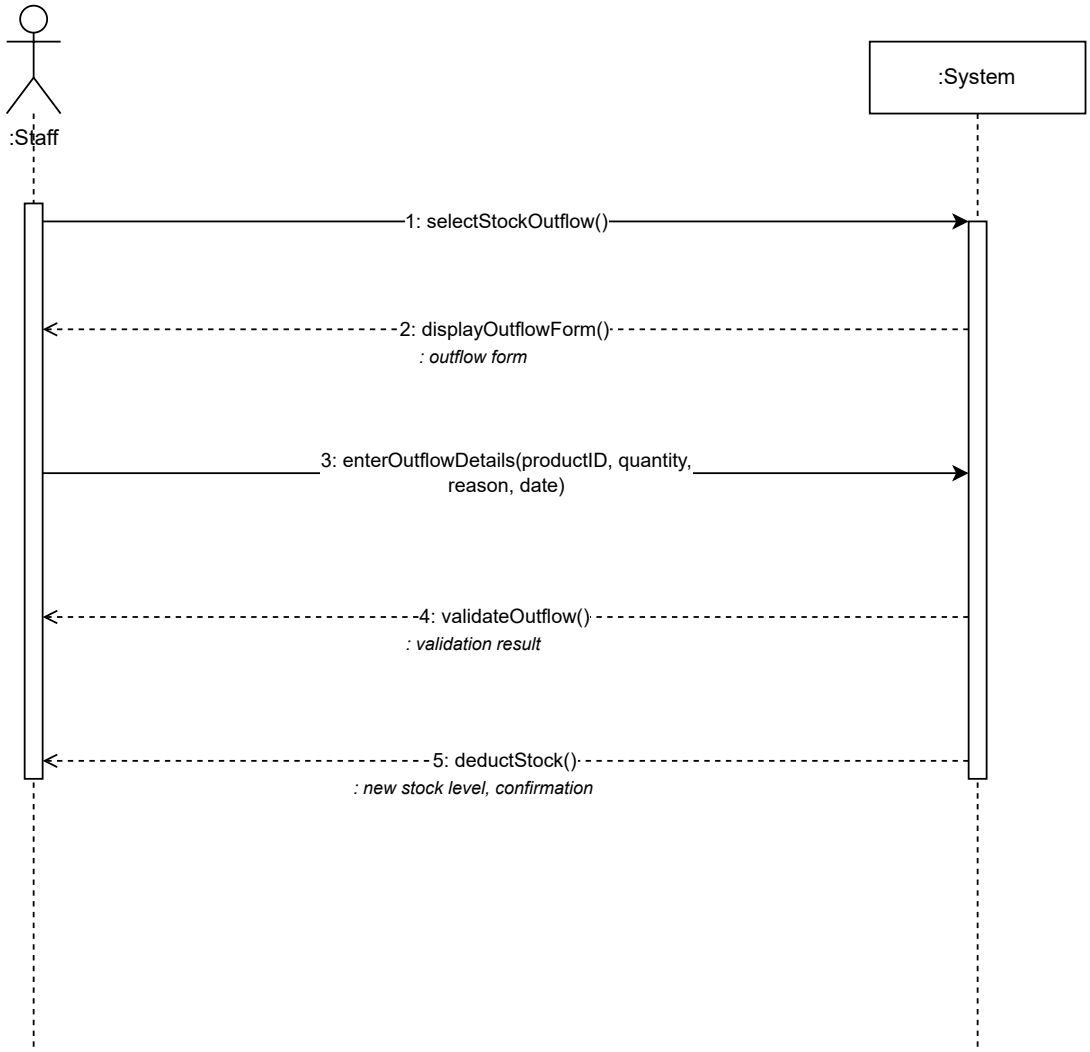
SSD - Update Product Details



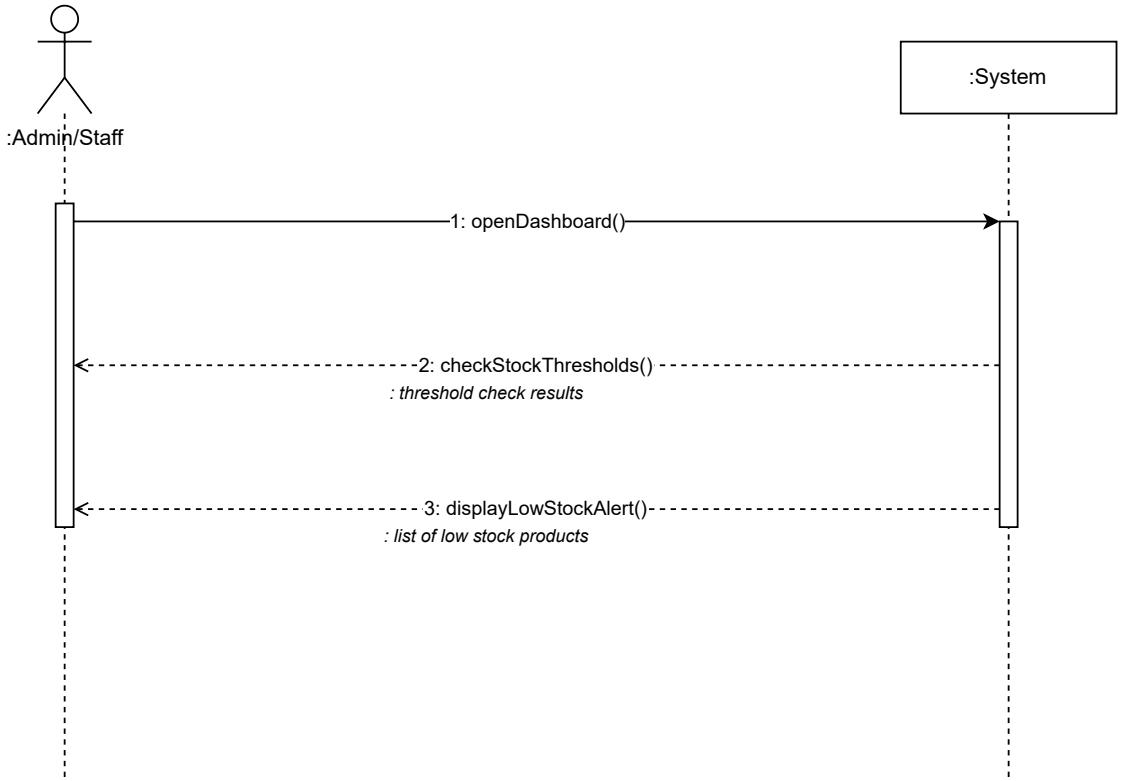
SSD - Record Stock Inflow



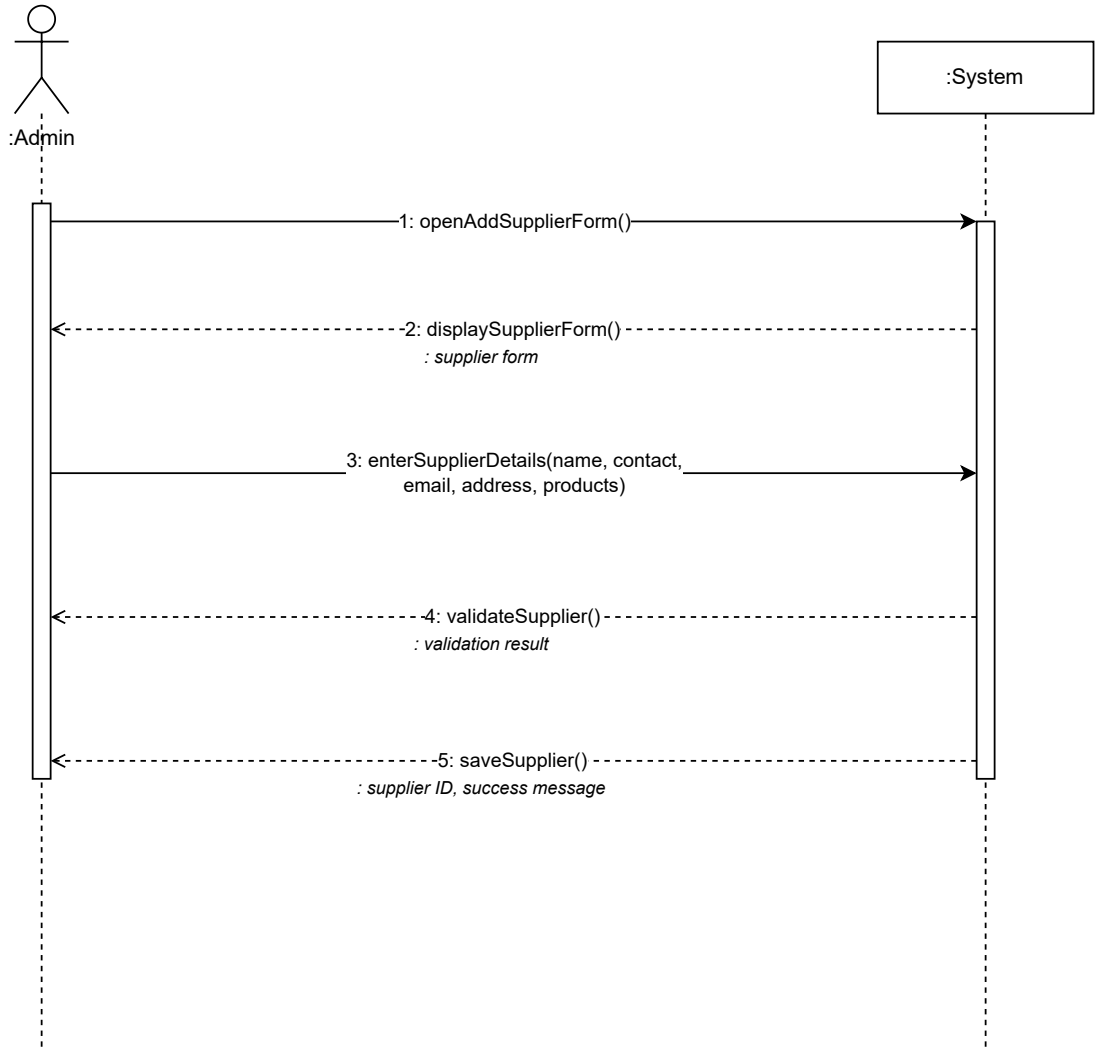
SSD - Process Stock Outflow



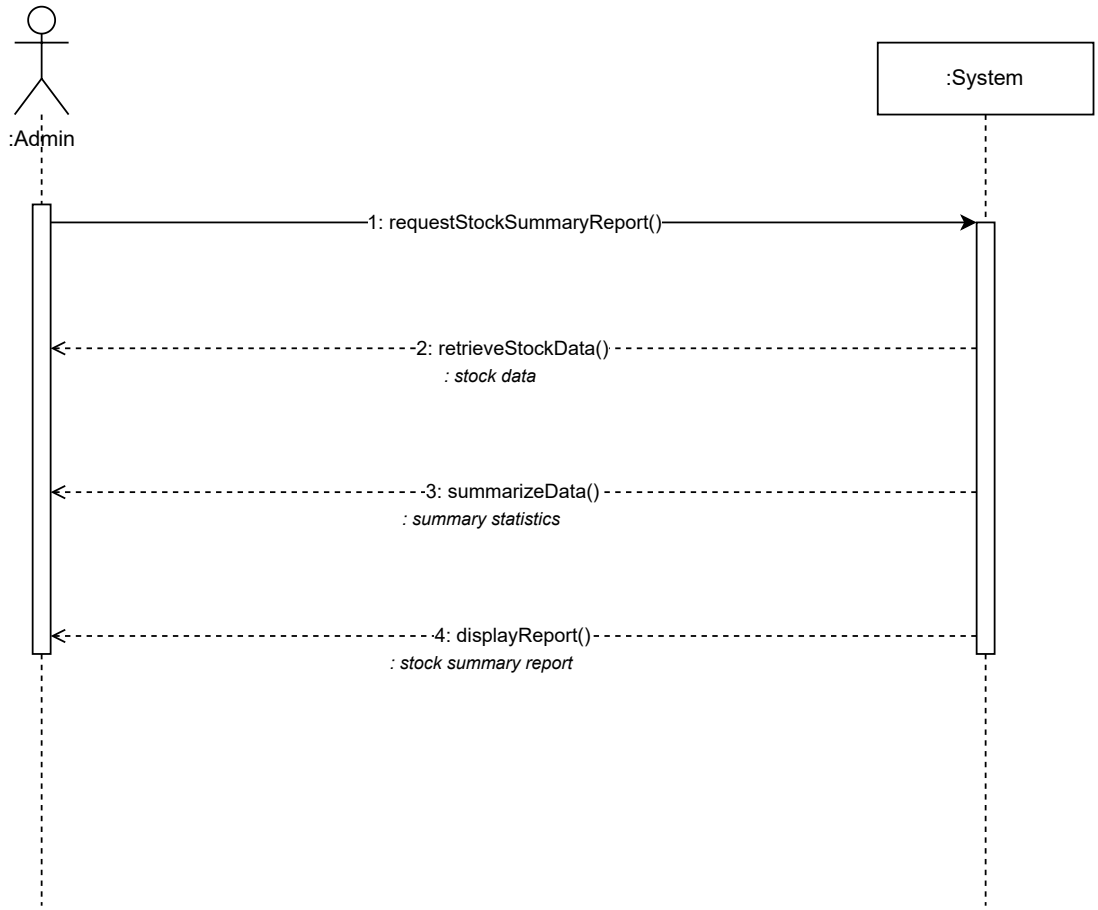
SSD - Check Low Stock Alerts



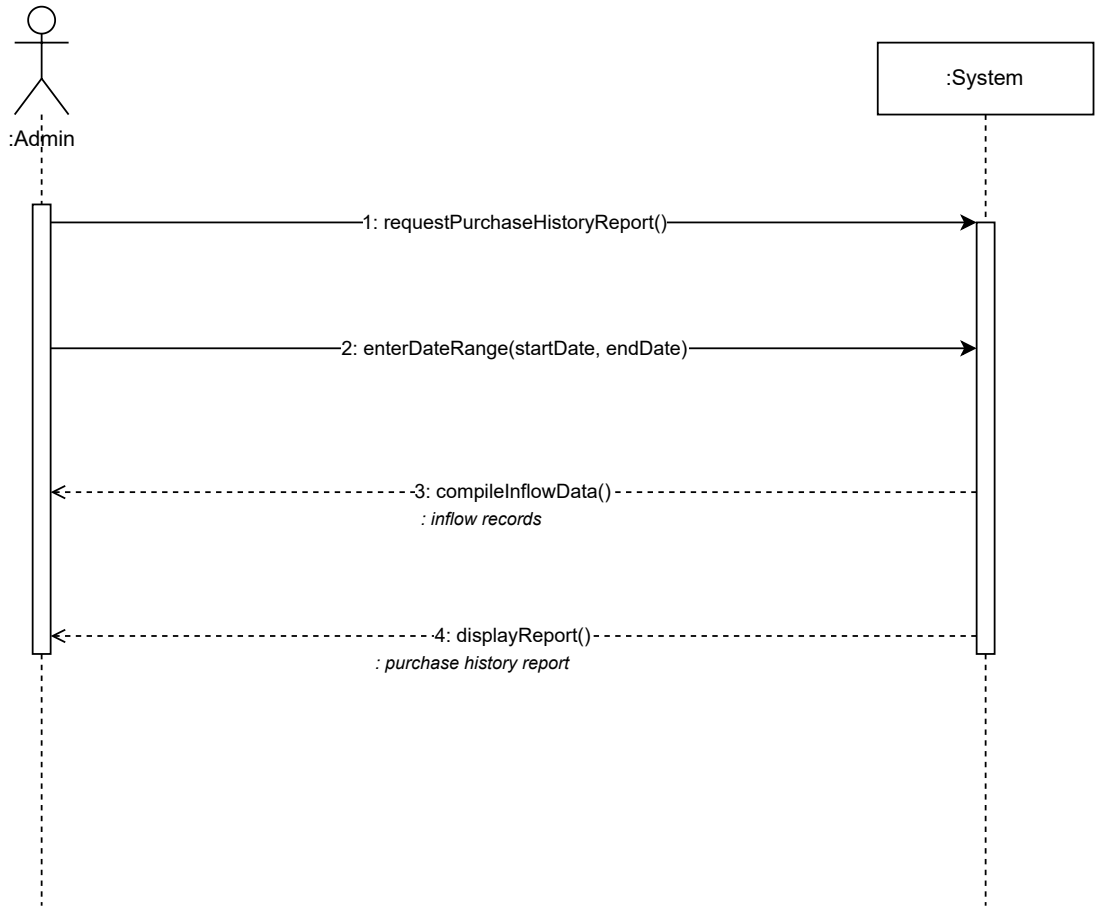
SSD - Add New Supplier



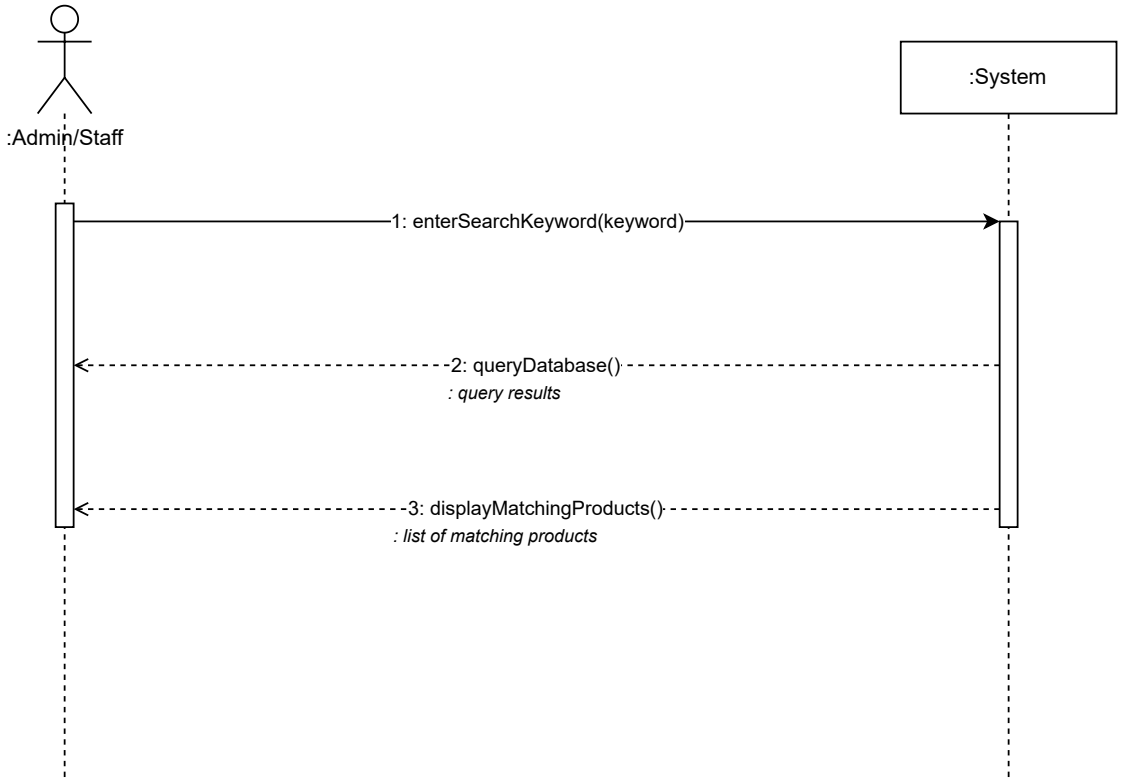
SSD - Generate Stock Summary Report



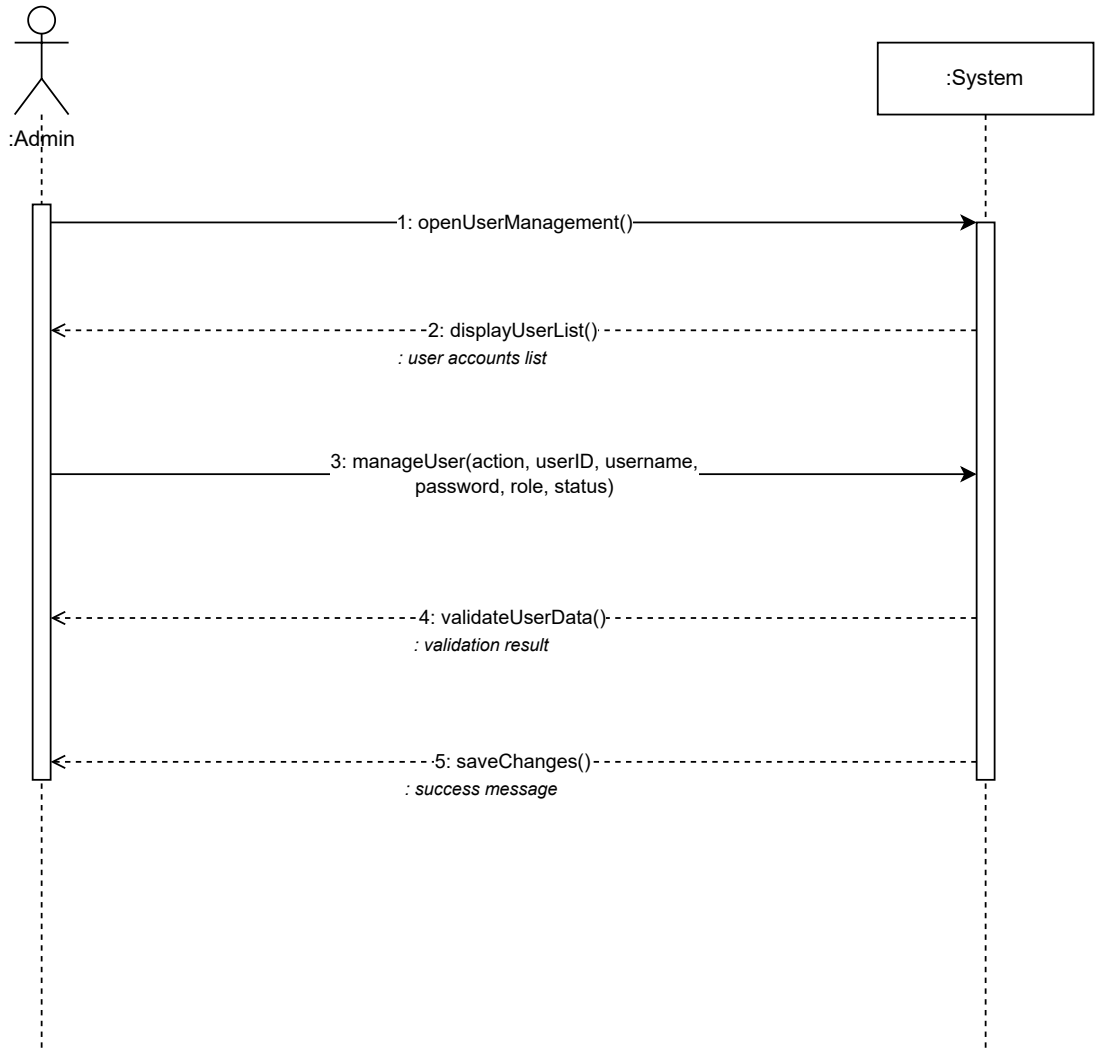
SSD - Generate Purchase History Report



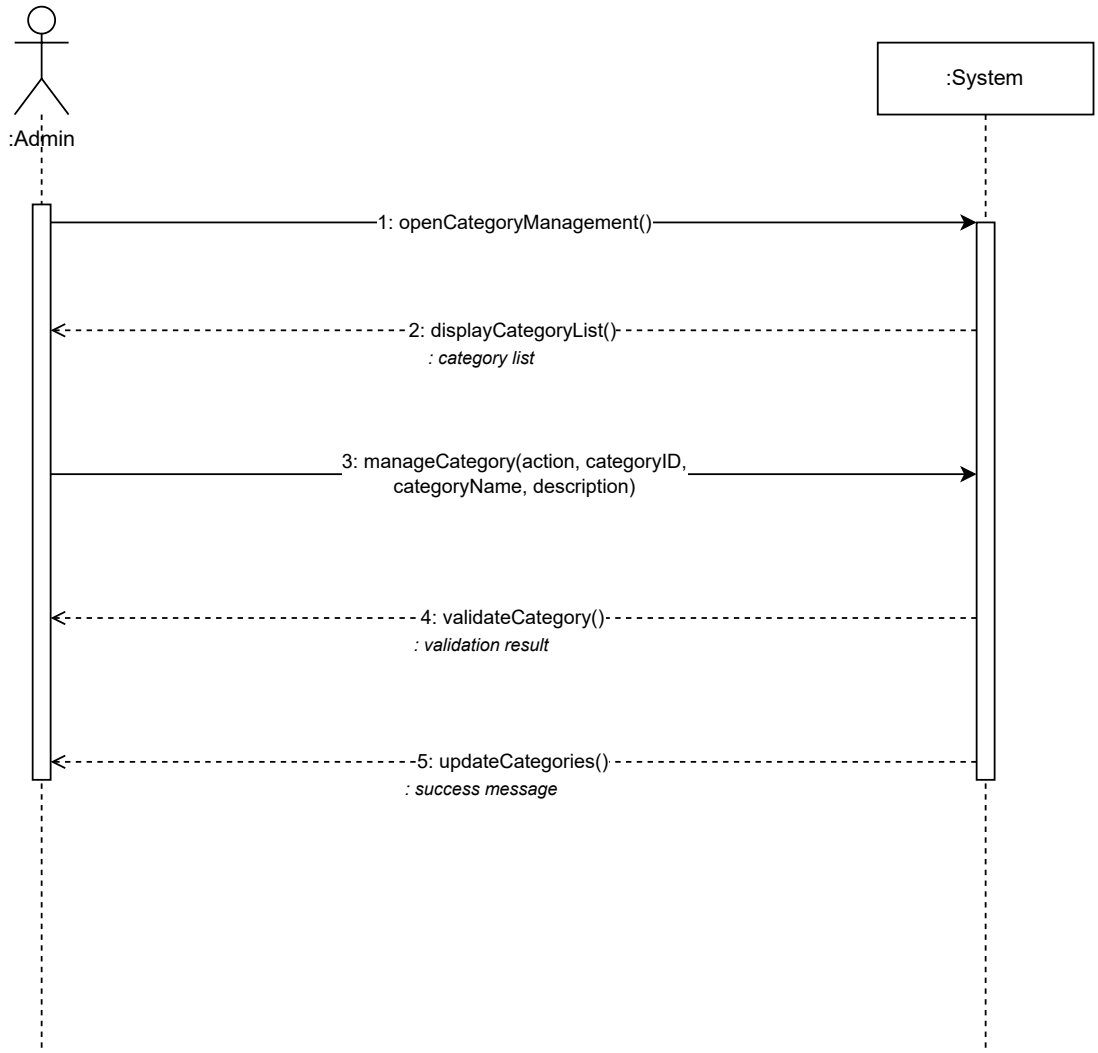
SSD - Search Product

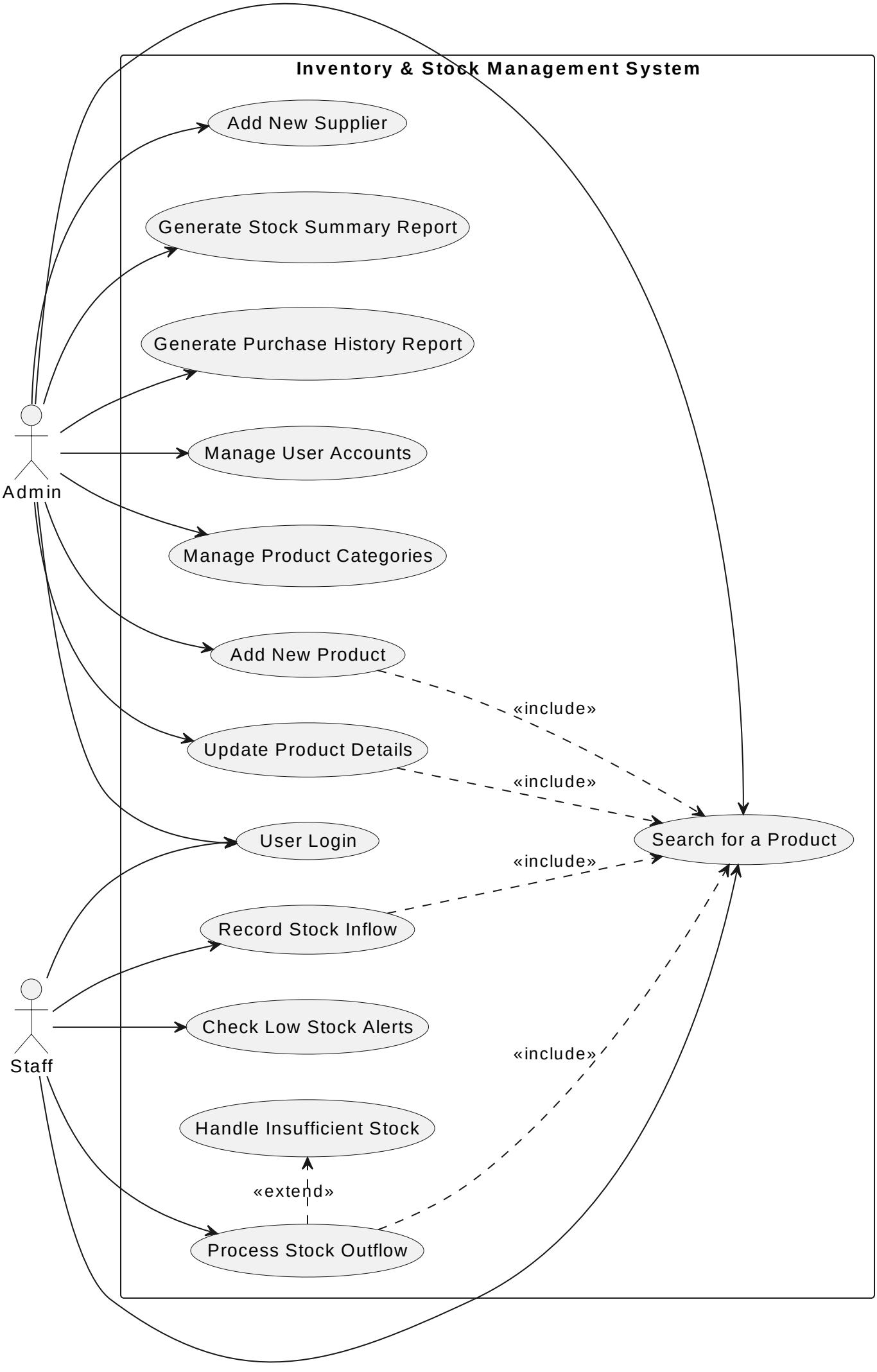


SSD - Manage User Accounts

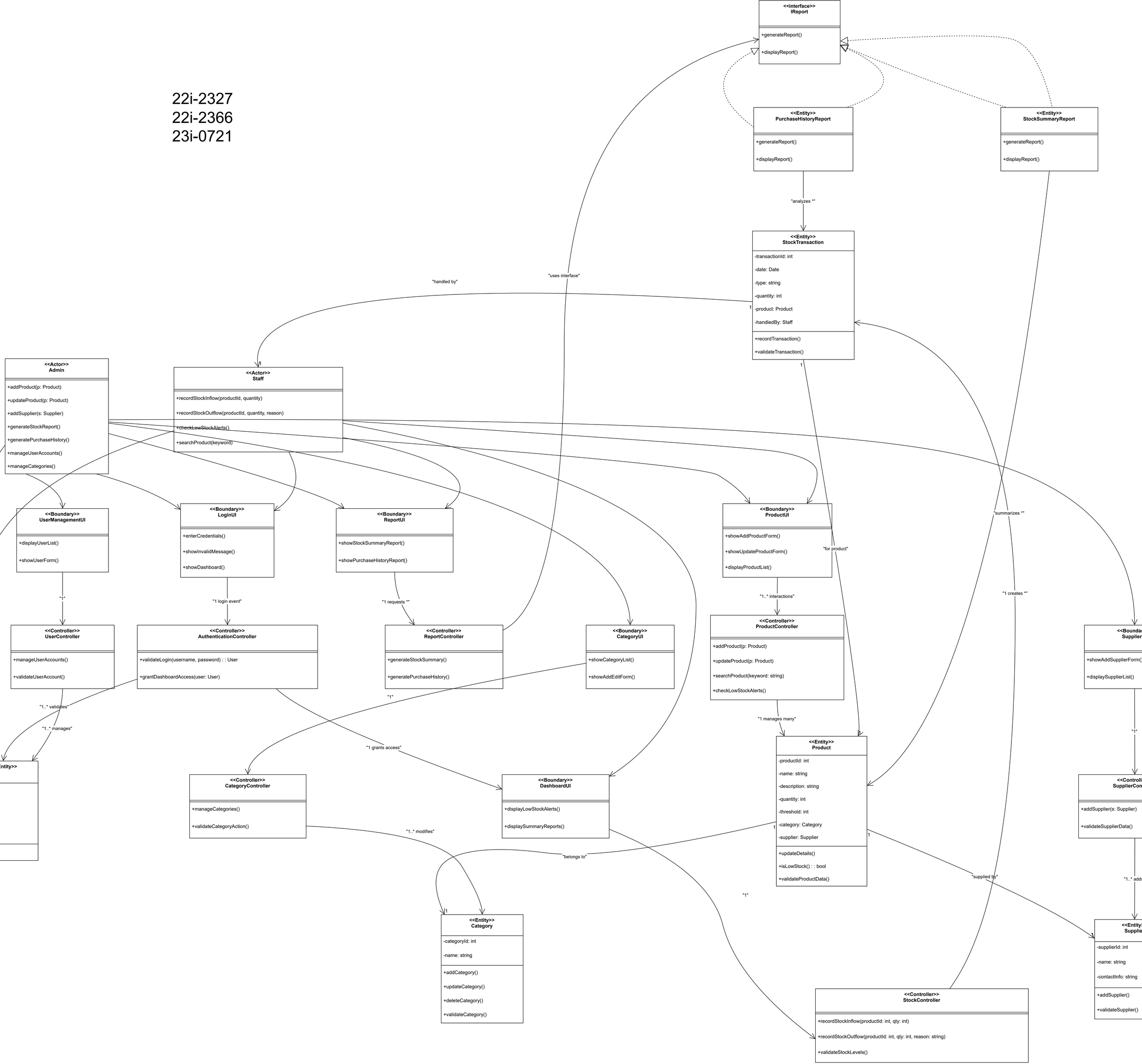


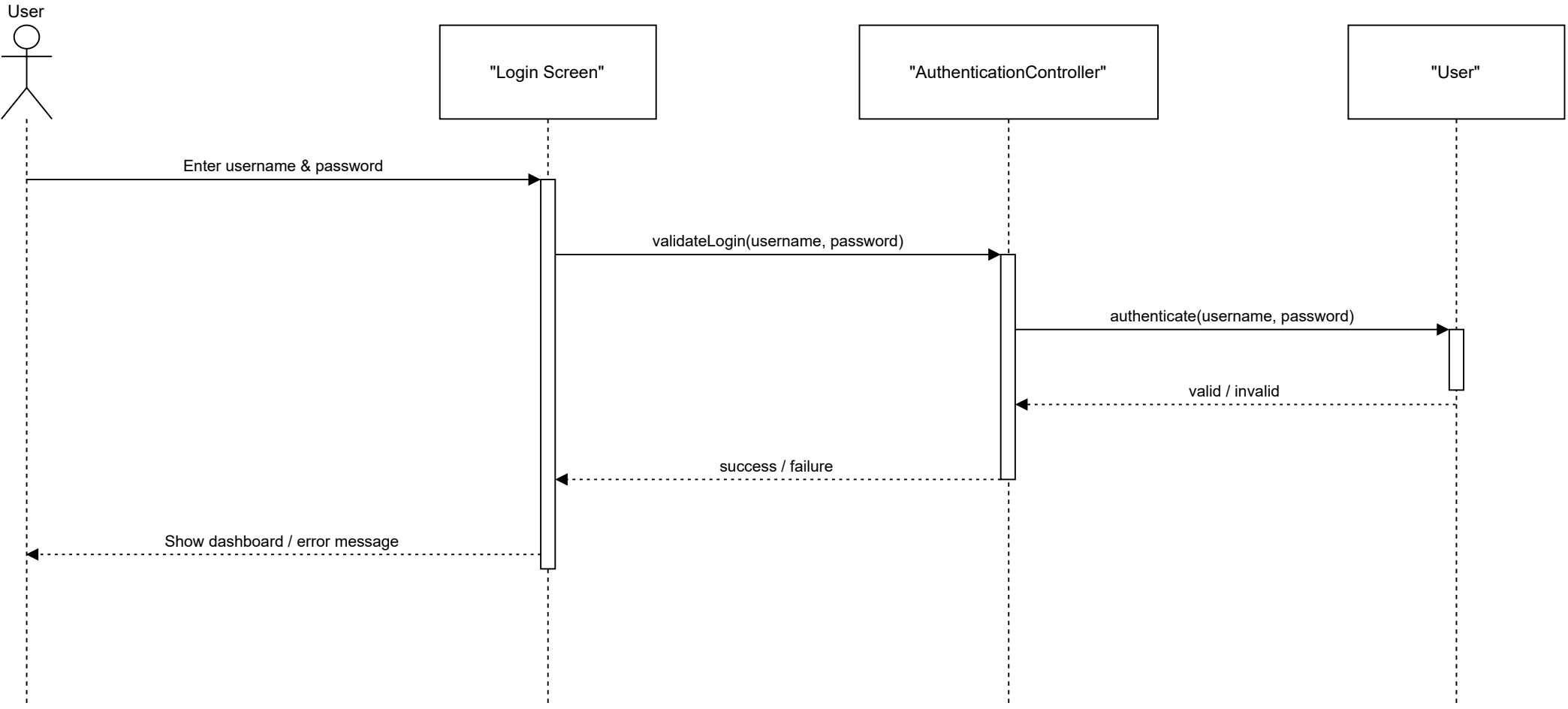
SSD - Manage Product Categories

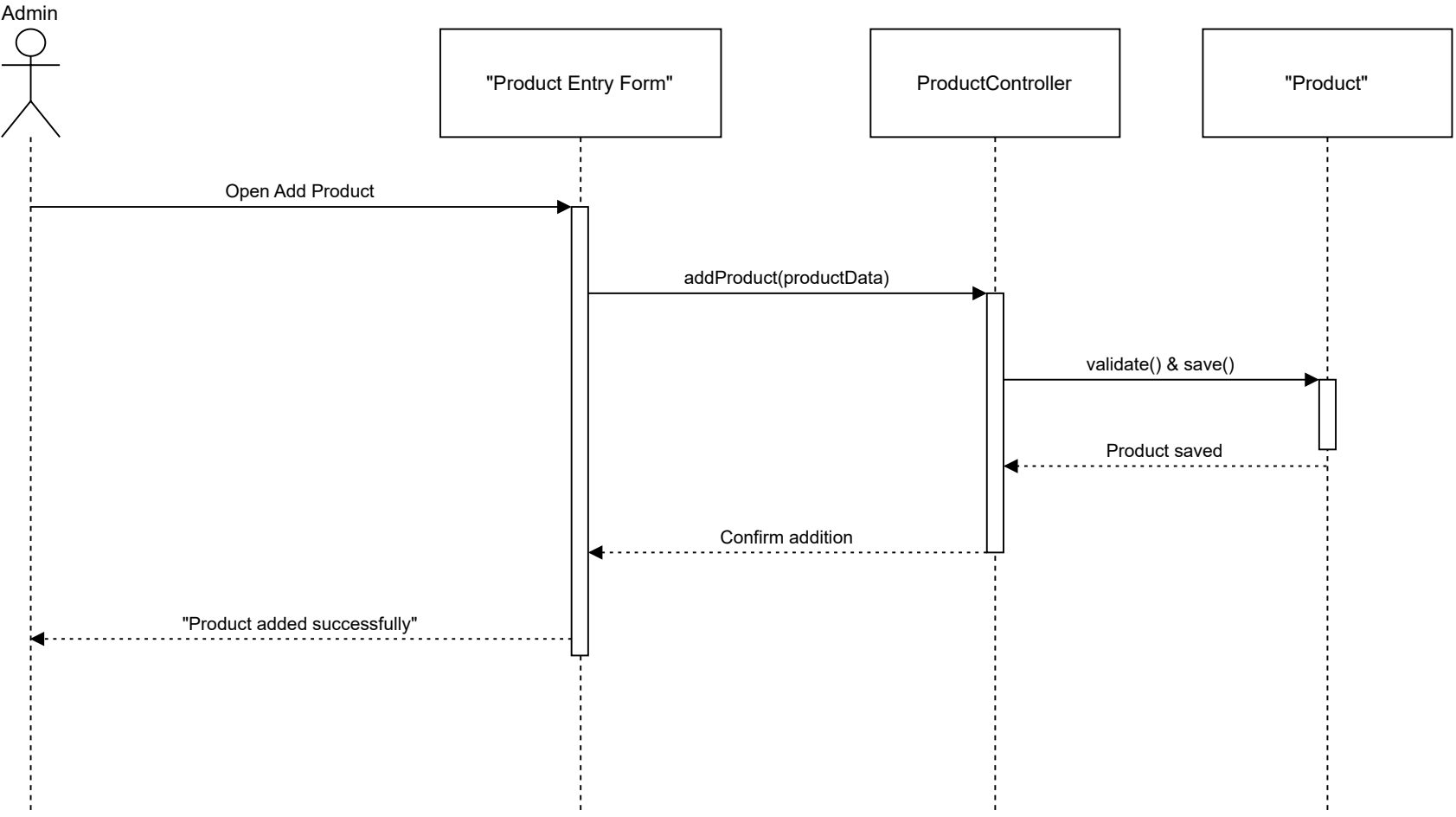


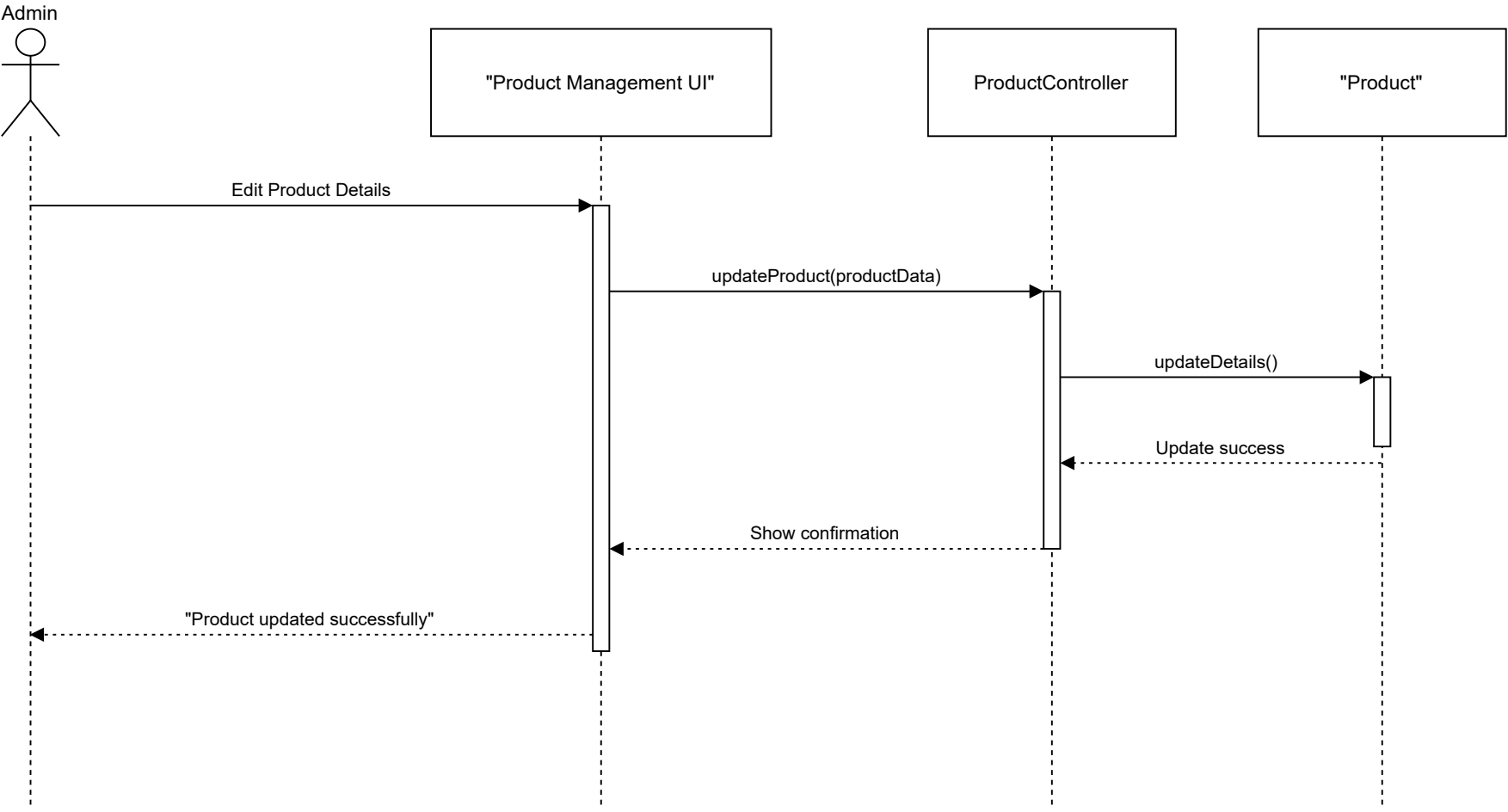

Inventory & Stock Management System
Add New Supplier
Generate Stock Summary Report
Generate Purchase History Report
Manage User Accounts
Manage Product Categories
Add New Product
Update Product Details
User Login
Search for a Product
Record Stock Inflow
Check Low Stock Alerts
Handle Insufficient Stock
Process Stock Outflow
Admin
Staff
«include»
«include»
«include»
«include»
«extend»

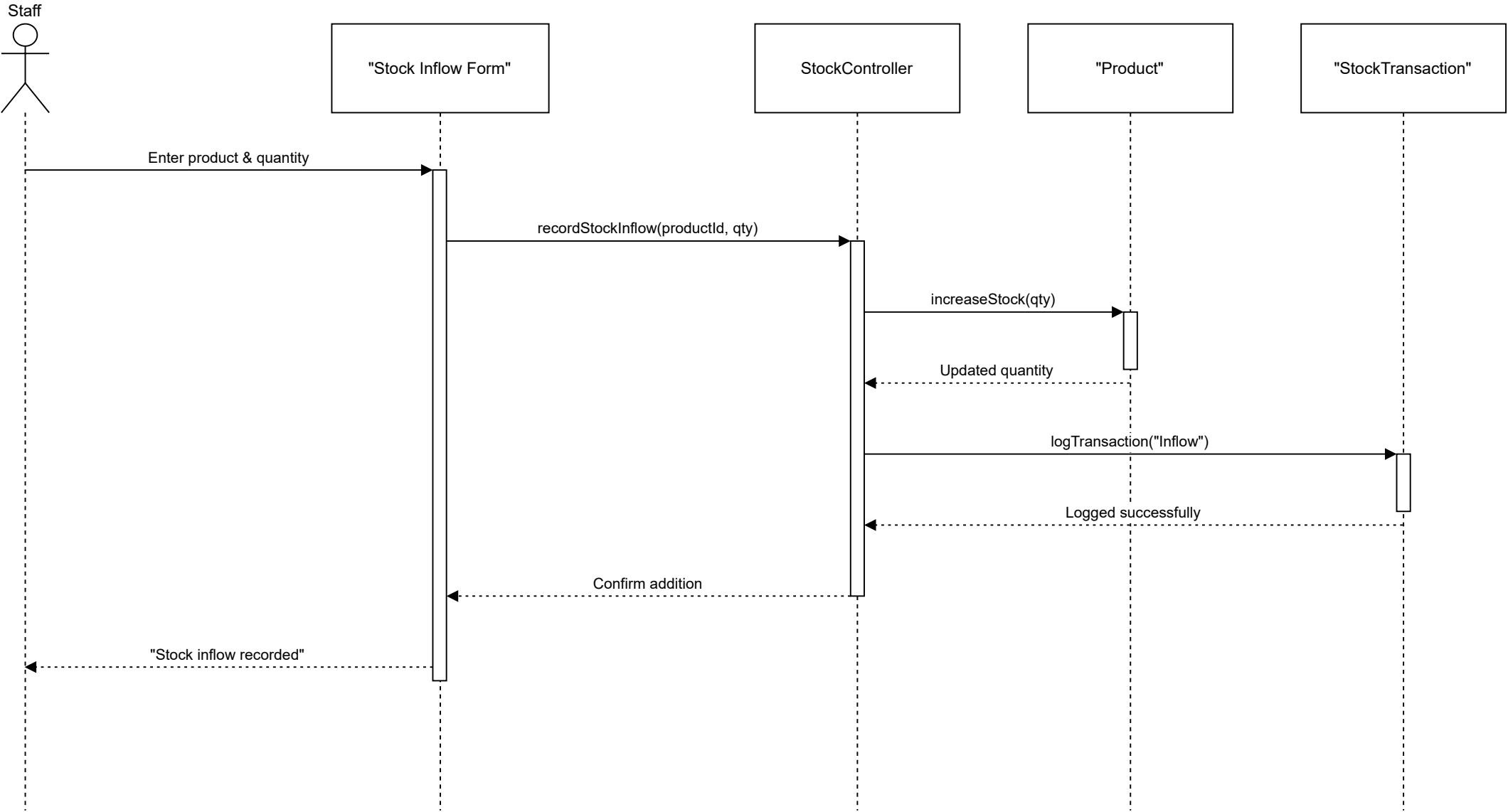
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22i-2366
23i-0721

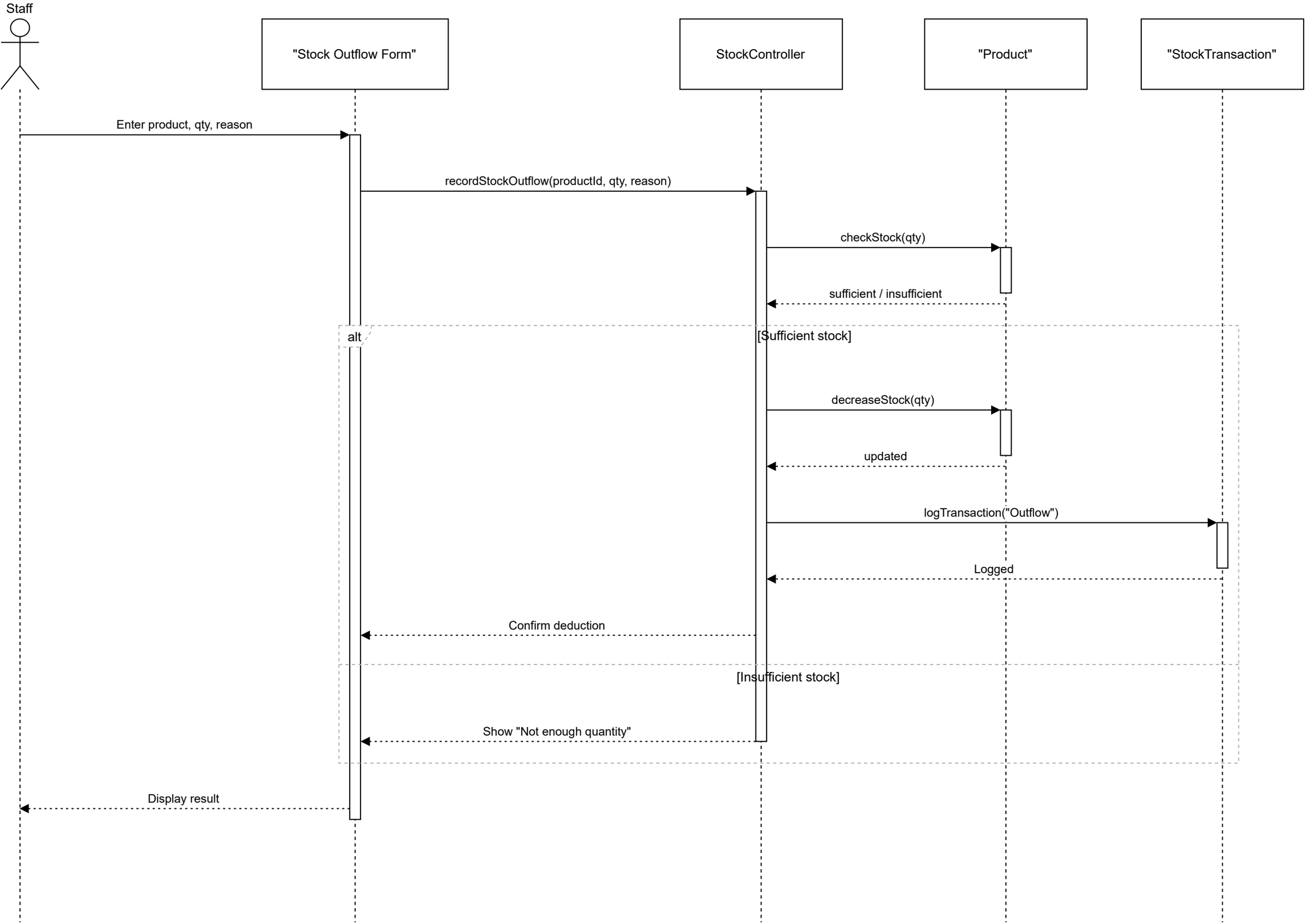


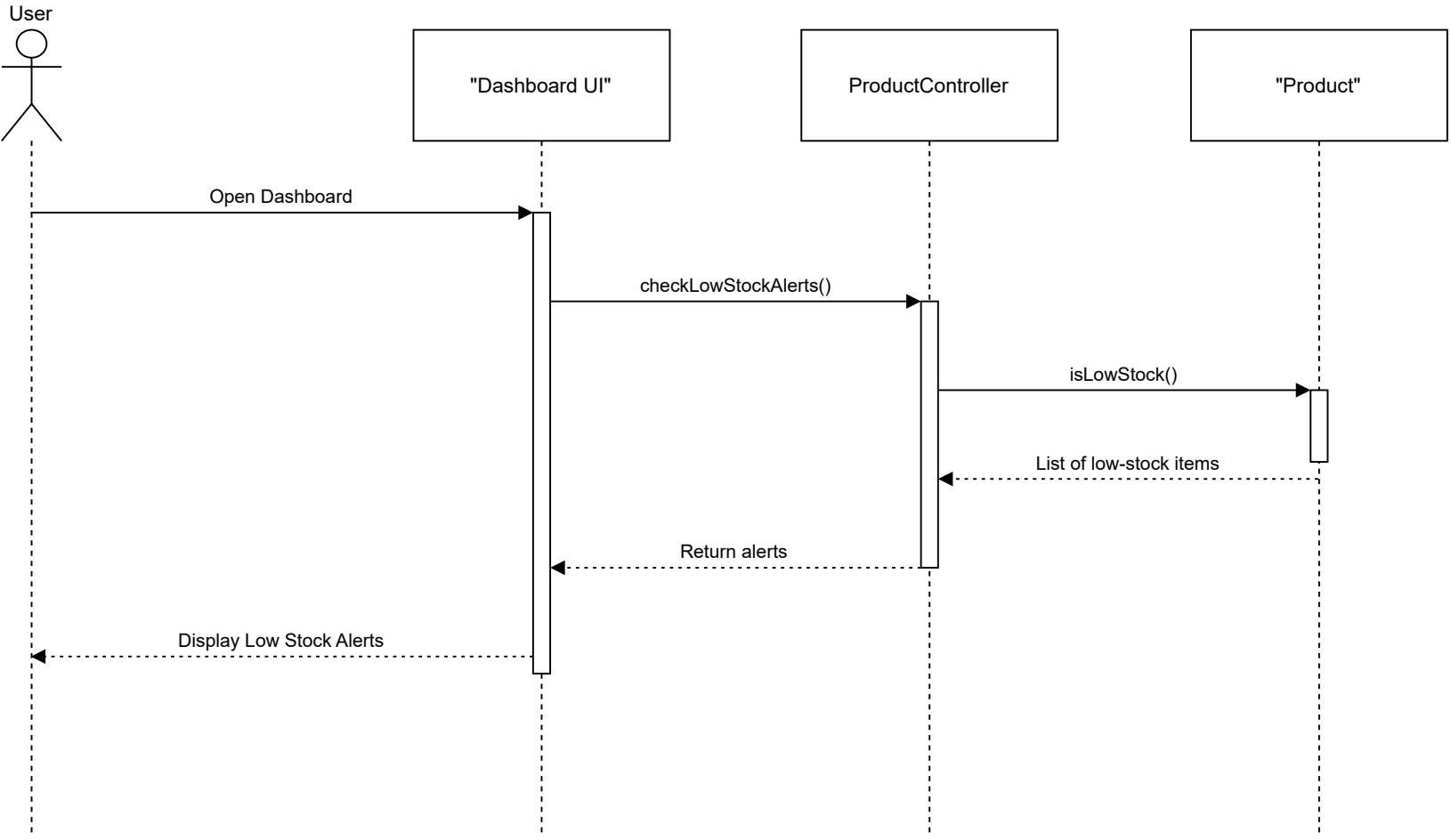

User
"Login Screen"
"AuthenticationController"
"User"
Enter username & password
validateLogin(username, password)
authenticate(username, password)
valid / invalid
success / failure
Show dashboard / error message

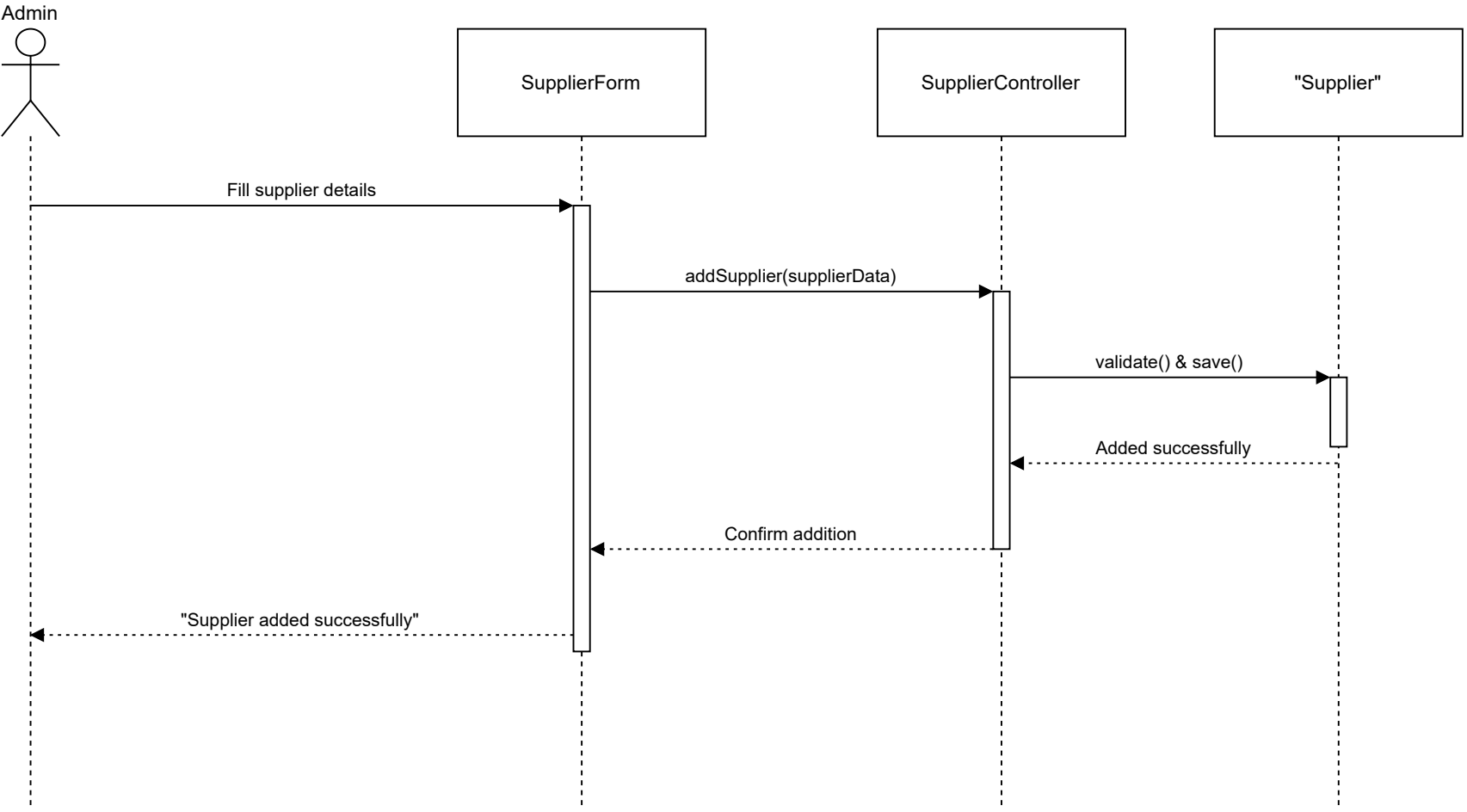

Admin
"Product Entry Form"
ProductController
"Product"
Open Add Product
addProduct(productData)
validate() & save()
Product saved
Confirm addition
"Product added successfully"

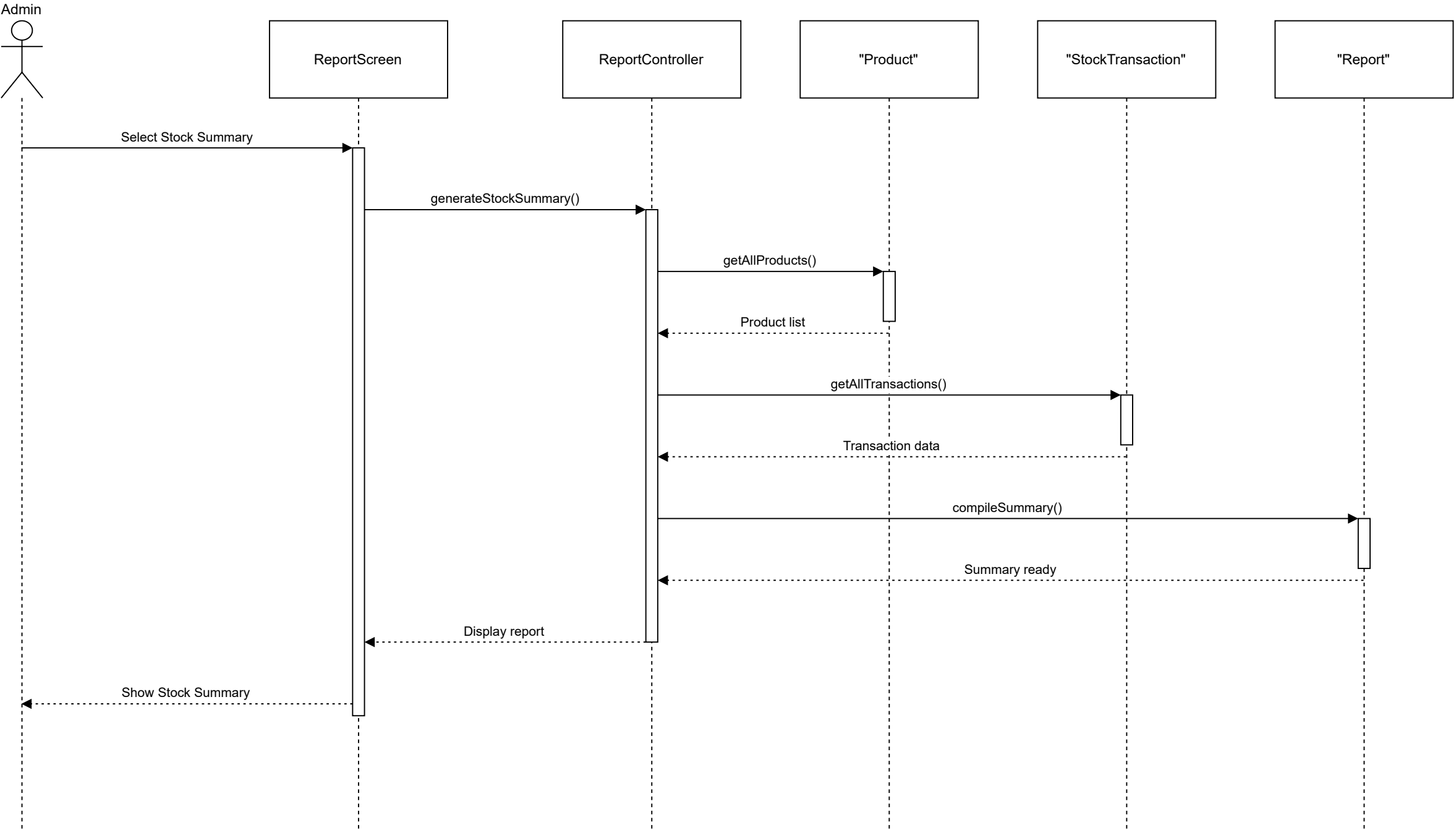

Admin
"Product Management UI"
ProductController
"Product"
Edit Product Details
updateProduct(productData)
updateDetails()
Update success
Show confirmation
"Product updated successfully"

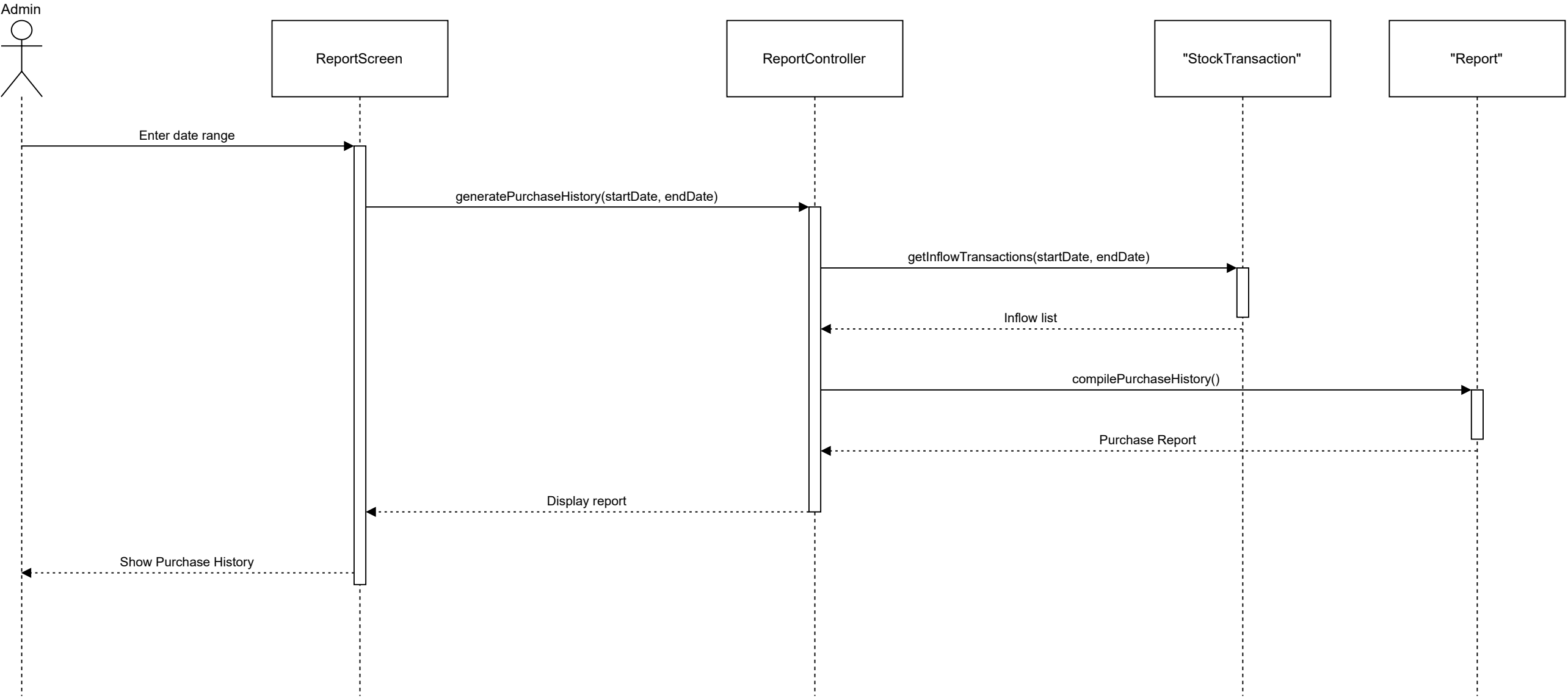

Staff
"Stock Inflow Form"
StockController
"Product"
"StockTransaction"
Enter product & quantity
recordStockInflow(productId, qty)
increaseStock(qty)
Updated quantity
logTransaction("Inflow")
Logged successfully
Confirm addition
"Stock inflow recorded"

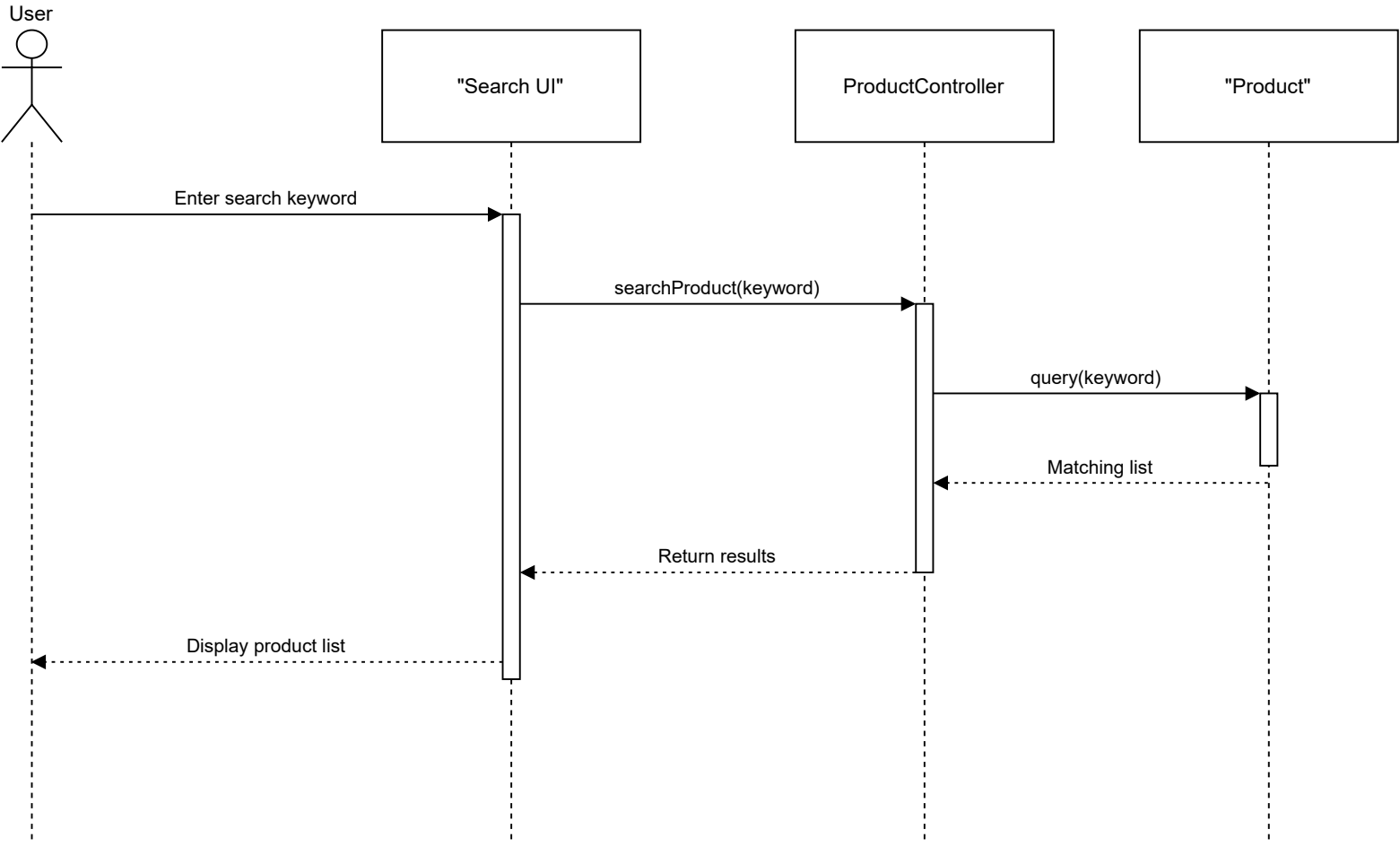

Staff
"Stock Outflow Form"
StockController
"Product"
"StockTransaction"
Enter product, qty, reason
recordStockOutflow(productId, qty, reason)
checkStock(qty)
sufficient / insufficient
alt
[Sufficient stock]
decreaseStock(qty)
updated
logTransaction("Outflow")
Logged
Confirm deduction
[Insufficient stock]
Show "Not enough quantity"
Display result

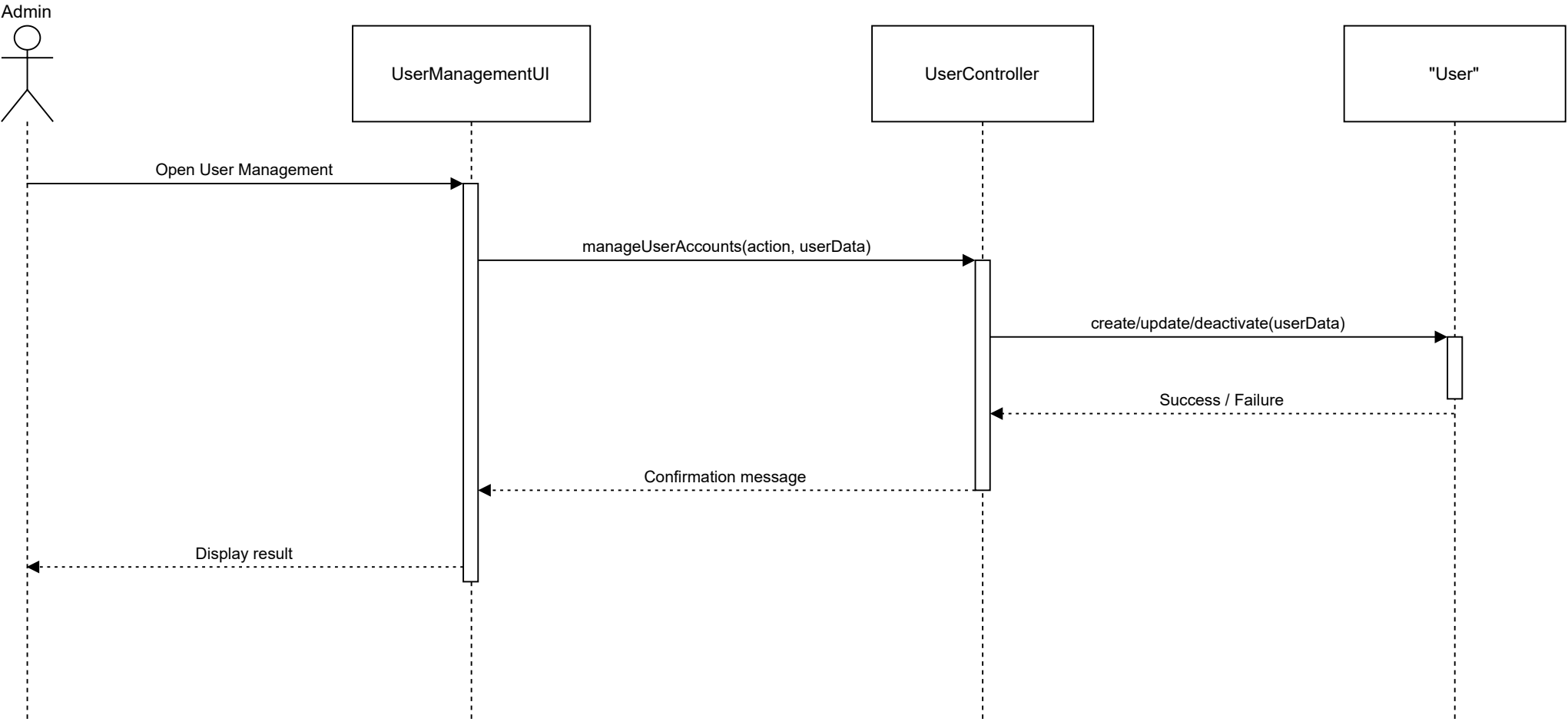

User
"Dashboard UI"
ProductController
"Product"
Open Dashboard
checkLowStockAlerts()
isLowStock()
List of low-stock items
Return alerts
Display Low Stock Alerts

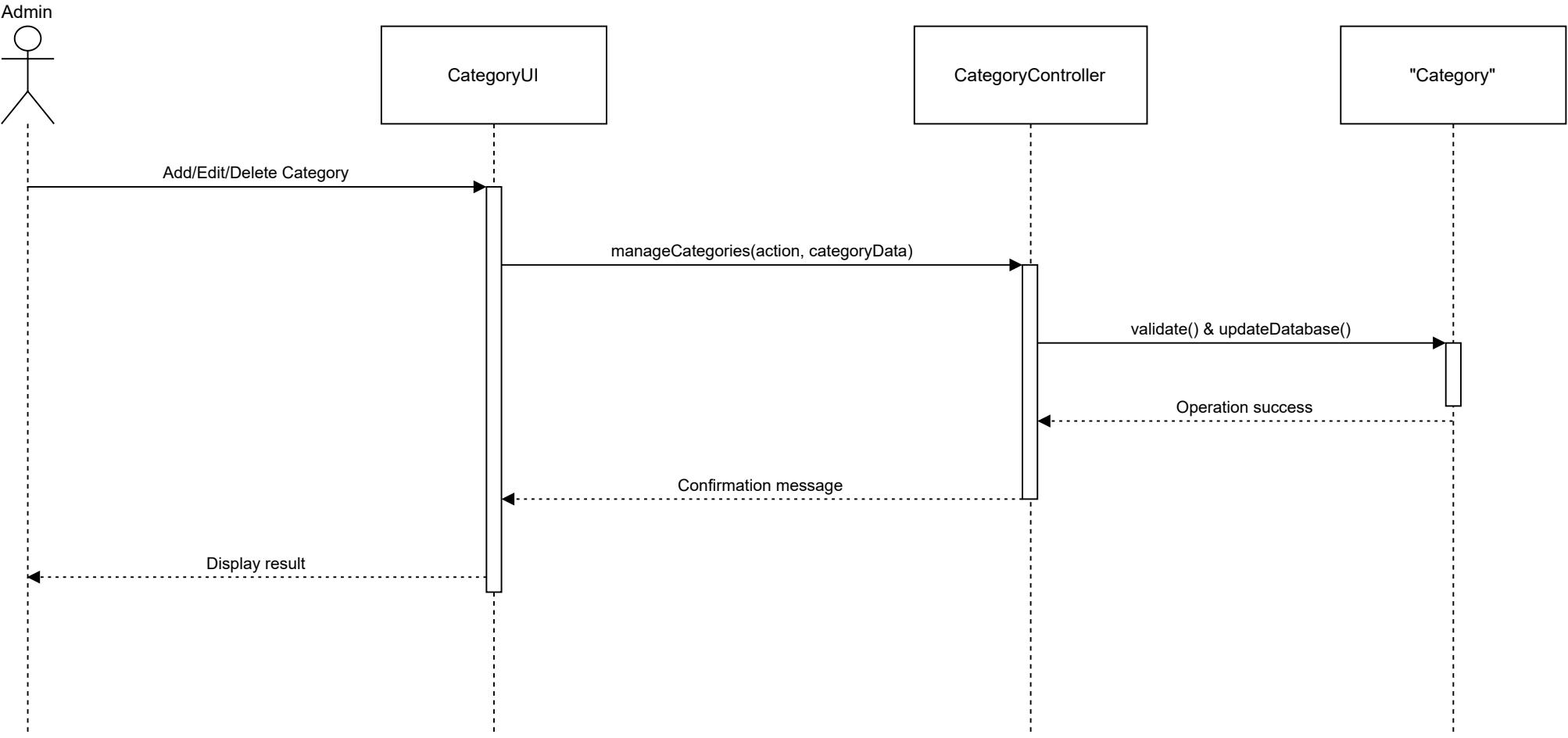

Admin
SupplierForm
SupplierController
"Supplier"
Fill supplier details
addSupplier(supplierData)
validate() & save()
Added successfully
Confirm addition
"Supplier added successfully"


Admin
ReportScreen
ReportController
"Product"
"StockTransaction"
"Report"
Select Stock Summary
generateStockSummary()
getAllProducts()
Product list
getAllTransactions()
Transaction data
compileSummary()
Summary ready
Display report
Show Stock Summary


Admin
ReportScreen
ReportController
"StockTransaction"
"Report"
Enter date range
generatePurchaseHistory(startDate, endDate)
getInflowTransactions(startDate, endDate)
Inflow list
compilePurchaseHistory()
Purchase Report
Display report
Show Purchase History


User
"Search UI"
ProductController
"Product"
Enter search keyword
searchProduct(keyword)
query(keyword)
Matching list
Return results
Display product list


Admin
UserManagementUI
UserController
"User"
Open User Management
manageUserAccounts(action, userData)
create/update/deactivate(userData)
Success / Failure
Confirmation message
Display result


Admin
CategoryUI
CategoryController
"Category"
Add/Edit/Delete Category
manageCategories(action, categoryData)
validate() & updateDatabase()
Operation success
Confirmation message
Display result